

An architectural capacity ceiling, not a barren plateau: why a fixed-encoding variational quantum circuit cannot fit the Lorenz-63 attractor

Tushar Pandey 

Texas A&M University, College Station, TX 77843, USA

E-mail: tusharp@tamu.edu

Submitted to: *Mach. Learn.: Sci. Technol.*

Abstract. Variational quantum circuits train poorly on chaotic forecasting, a failure usually blamed on barren plateaus (exponentially vanishing gradients). Using an exactly simulable four-qubit variational quantum physics-informed circuit fit to the Lorenz-63 system, we show the barren-plateau explanation fails and identify the failure as an architectural capacity ceiling fixed by the circuit’s time-encoding, not its trainable depth. Four measurements support this. (i) A McClean-comparable gradient-variance estimator sits at the local-cost Haar/2-design scale $2^{-2n} = 3.9 \times 10^{-3}$ at $n = 4$; on its structurally live parameters it decays about ninefold with depth, then saturates there, large enough to train, not an exponential collapse to zero. (ii) At a common nominal budget of 200 optimiser iterations (600, in three stages, for layer-wise), three optimiser families (gradient descent, layer-wise, SPSA) reach the same order of magnitude of loss, with finite-difference and exact parameter-shift gradients agreeing to about 2×10^{-5} relative as a consistency check, so no optimiser unlocks a better basin. (iii) The output-Jacobian rank saturates at 33 from five layers on, so depth buys no new output directions. (iv) A Fourier analysis explains why: the qubit-1 phase encoding acts on the initial $|0\rangle$ and is inert, so the maximum accessible frequency is $2.5/t_{\max} \approx 0.83$ Hz, identical at every depth and a factor of about 4.4 below the narrowest Lorenz component’s bandwidth. The corrected band has dimension $1 + 2 \times 5 = 11$ per observable, and $3 \times 11 = 33$ equals the measured rank ceiling exactly, unifying the two diagnostics. A trained depth sweep agrees: mean loss improves with depth, then flattens once the rank saturates. We correct our earlier preprint diagnosis, which compared unnormalised gradient norms to the McClean threshold, and place the advantage of fixed reservoirs and classical echo-state networks in architecture, not quantum mechanics.

Keywords: variational quantum circuits, barren plateaus, quantum machine learning, Fourier expressivity, reservoir computing, physics-informed neural networks

1. Introduction

A recurring empirical observation in quantum machine learning is that variational quantum circuits, trained end-to-end by gradient descent, struggle to fit hard temporal targets such as chaotic trajectories [1, 2, 7]. When such a model fails to reach a useful loss, the default explanation in the literature is the barren plateau: for sufficiently expressive random circuits, the variance of the loss gradient with respect to any single parameter vanishes exponentially in the qubit count or the circuit depth, so that the loss landscape becomes an exponentially flat plateau on which gradient descent stalls [1, 2].

Testing that diagnosis requires care about scale. At fixed qubit count an absolute-threshold statement is not available: the McClean bound is a scaling statement in the qubit number n , and a single small n cannot be used to assert that a variance has or has not fallen below an n -asymptotic threshold. What a single n can show is whether the gradient variance decays with circuit size and where it sits relative to the Haar/2-design saturation scale for the same ansatz and cost. That comparison must be made with a normalised, dimensionless quantity: an absolute gradient magnitude carries the arbitrary scale of the loss function and cannot be compared to a 2^{-n} threshold at all. Neglecting this normalisation is precisely the error we correct here relative to our own earlier preprint (Section 5.1).

This paper studies a deliberately small and exactly simulable case: a four-qubit variational quantum physics-informed neural network (QPINN) asked to fit the Lorenz-63 attractor [8, 9, 10, 11]. Small scale is deliberate. At four qubits every diagnostic we need can be computed exactly, with exact parameter-shift gradients and full statevector Jacobians, so no statistical or sampling artefact can be mistaken for a barren plateau. We make no claim about near-term hardware relevance; the contribution is a mechanistic diagnosis, not a device demonstration.

Contribution. We show that the QPINN failure on Lorenz-63 is not a barren plateau but an architectural capacity ceiling set by the circuit’s fixed time-encoding. The evidence is four measurements (Section 4):

- (i) a McClean-comparable gradient-variance estimator that sits at the local-cost Haar/2-design floor and, on its live parameters, decays toward that floor with depth rather than collapsing to zero (Section 4.1);
- (ii) optimiser-independence of the final loss at matched iteration budget (Section 4.2);
- (iii) saturation of the output-Jacobian effective dimension with depth (Section 4.3);
- (iv) a depth-independent Fourier-expressivity ceiling below the target bandwidth (Section 4.4), whose band dimension equals the measured rank ceiling exactly (Section 4.5),

consistent with a trained depth sweep whose mean loss is flat once the Jacobian rank has saturated (Section 4.6). We then correct our earlier gradient-norm diagnosis (Section 5.1) and argue that the observed advantage of a fixed reservoir, and of a

classical echo-state network, over the variational circuit is architectural, not quantum (Section 5.3).

2. Related work

Barren plateaus and trainability. McClean et al. [1] established that random parameterised quantum circuits exhibit exponentially concentrating expectation values and hence exponentially small gradient variance, and Cerezo et al. [2] surveyed the resulting trainability landscape and its distinct causes (expressibility-induced plateaus, cost-function locality, entanglement-induced plateaus, and noise-induced plateaus). Cost-function locality matters here: for a local cost the Haar/2-design variance floor is $\Theta(2^{-2n})$ rather than $\Theta(2^{-n})$ [3], and our diagnostic cost is local. Beyond the original global-cost analysis, the modern literature distinguishes several trainability phenomena, including local barren plateaus, layer-wise trainability, and observable-dependent concentration [3, 4, 5]. Our live/dead parameter split (Section 3.2) is exactly an observable-locality analysis of this kind: the local cost reads only $\langle Z_0 \rangle$, so at shallow depth many parameters sit outside the backward light cone of Z_0 and have identically zero gradient, and we report the live variance separately rather than diluting it with these structural zeros. On the capacity side, expressivity- and approximation-theoretic tools have been used to enlarge the reachable function class, for example tensor-network-guided hypernetworks for variational circuits [6]; our diagnosis is complementary, showing that for a fixed encoding the reachable class is band-limited independently of the trainable block. A central point of the present paper is that these trainability pathologies are diagnostically distinct from a capacity shortfall, and that the two must be separated with a properly normalised measurement.

Expressivity of encodings. Schuld, Sweke, and Meyer [12] showed that a data-encoding quantum circuit realises a truncated Fourier series in the input, whose accessible frequency spectrum is fixed by the data-encoding gates, while the trainable blocks only set the Fourier coefficients within that spectrum. Our Fourier diagnostic (Section 4.4) is a direct application of this result: the maximum accessible frequency is a property of the fixed encoding and is therefore independent of variational depth.

Reservoir computing, classical and quantum. Reservoir computing [13, 14] fixes a high-dimensional nonlinear dynamical system and trains only a linear readout, and quantum reservoir computing [15, 16, 17, 18] transplants this to a fixed quantum map. Our companion fair-baseline benchmarking study [20] shows that on chaotic forecasting a tuned classical echo-state network matches or beats a quantum reservoir at comparable feature scale, and a companion conditioning analysis [19] traces the reservoir advantage to the bounded condition number of a fixed random feature map relative to an end-to-end-trained one. The present paper is the trainability-side complement: it explains why the variational alternative, specifically, is capacity-limited on this task.

Relation to our prior work. The base QPINN/QRC architecture and the Lorenz forecasting setup were introduced in [7], which also reports the QPINN-versus-reservoir performance gap we build on here; we reuse that setup and do not re-derive the circuits beyond what is needed for self-containment. The diagnosis reported here supersedes the trainability claim of an earlier, unpublished version of this manuscript, whose gradient-norm comparison was not normalised to the McClean threshold (Section 5.1).

3. Methods

3.1. Circuit and task

We reuse the four-qubit QPINN and the Lorenz-63 forecasting task of [7], summarised here for self-containment. Time is normalised to $[0, 2\pi]$ and encoded by single-qubit rotations,

$$\theta_t = \frac{2\pi t}{t_{\max}}, \quad U_{\text{enc}}(t) = R_Y^{(0)}(\theta_t) R_Z^{(1)}(\theta_t) R_X^{(2)}(\theta_t) R_Y^{(3)}(\theta_t/2), \quad (1)$$

where the fourth qubit is driven at half the base frequency. Each of L variational layers applies $R_X R_Y R_Z$ rotations on every qubit followed by a CNOT entangling chain with R_Z phases on the targets, giving $15L$ trainable parameters. The physical state estimate is a linear map of the Pauli- Z expectations $\langle Z_0 \rangle, \langle Z_1 \rangle, \langle Z_2 \rangle$. The physics-informed loss combines the Lorenz ODE residual at $N_c = 50$ collocation times with an initial-condition term,

$$\mathcal{L}_\theta = \frac{1}{N_c} \sum_{i=1}^{N_c} \|\partial_t f_\theta(t_i) - F(f_\theta(t_i))\|^2 + \lambda \|f_\theta(0) - x_0\|^2, \quad (2)$$

with $\lambda = 10$ and F the Lorenz vector field ($\sigma = 10, \rho = 28, \beta = 8/3$). All quantities below are computed by exact statevector simulation on $n = 4$ qubits.

One structural feature of this encoding is load-bearing for the Fourier diagnostic and is worth stating up front. The qubit-1 encoding gate $R_Z^{(1)}(\theta_t)$ is the first gate acting on that qubit, which starts in $|0\rangle$; an R_Z rotation on a computational-basis state contributes only a global phase, so the qubit-1 encoding frequency is inert and never enters any observable. We confirm this numerically in Section 4.4: deleting the gate changes $\langle Z_0 \rangle, \langle Z_1 \rangle, \langle Z_2 \rangle$ by at most 1.6×10^{-15} at every depth. The consequence is that the accessible spectrum is set by three active encoding frequencies, not four (Section 3.5).

3.2. McClean-comparable gradient variance

The barren-plateau prediction is a statement about the variance, across random initialisations, of the gradient of a fixed circuit (a fixed input) with respect to a single parameter. It is essential that no averaging over the input (here, over time) is performed before differentiating, because time-averaging the cost annihilates all non-DC Fourier modes of the encoding and destroys exactly the quantity the McClean bound concerns.

We define the McClean-comparable estimator on the bounded, local cost $C_t = \frac{1}{2}(1 - \langle Z_0 \rangle(t)) \in [0, 1]$ at each fixed collocation time t . For each of 100 random initialisations we compute the exact gradient $\partial C_t / \partial \theta_k$ by the two-evaluation $\pm\pi/2$ parameter-shift rule (no finite differencing). Because each t is a fixed circuit and the cost is bounded in $[0, 1]$, the across-initialisation variance $\text{Var}_{\text{init}}[\partial_{\theta_k} C_t]$ is dimensionless and directly comparable to a 2^{-n} -type scale. We sweep $L \in \{1, 3, 5\}$ (that is, 15, 45, 75 parameters) on 24 fixed collocation times.

The cost C_t reads only $\langle Z_0 \rangle$, and the CNOT entangler flows one-way along $0 \rightarrow 1 \rightarrow 2 \rightarrow 3$. In the Heisenberg picture Z_0 therefore commutes with the entire entangler and with every last-layer gate except $R_X^{(0)}, R_Y^{(0)}$, so at low depth a large fraction of parameters have identically zero gradient: they lie outside the backward light cone of Z_0 . We report the pooled mean variance $\text{Var}_{\text{est}} = \langle \text{Var}_{\text{init}}[\partial_{\theta_k} C_t] \rangle_{k,t}$, but we also report it separately for the structurally live and dead parameters, because pooling over a dead-parameter fraction that shrinks with depth confounds light-cone geometry with plateau physics. A parameter-time cell is counted dead when its across-initialisation variance is below 10^{-28} ; the maximum absolute gradient among dead cells is 3.5×10^{-16} , confirming these are exact structural zeros rather than small finite values.

3.3. Matched-budget optimiser comparison

To test whether the failure is an artefact of a single optimiser, we train the full physics loss with four training configurations at a common nominal budget of 200 optimiser iterations (with the layer-wise and SPSA exceptions detailed in the protocol caveats below): (a) forward finite-difference gradients ($\epsilon = 10^{-4}$, 10 seeds); (b) exact parameter-shift gradients propagated analytically through the quadratic loss (5 seeds); (c) a layer-wise schedule that optimises one variational layer at a time (5 seeds); and (d) SPSA (5 seeds). Sources (a)–(c) share the Adam meta-optimiser settings, learning rate, decay, and gradient-clipping percentile of [7]; SPSA instead uses standard Spall gain schedules ($\alpha = 0.602$, $\gamma = 0.101$, median-calibrated step size) with no Adam and no clipping.

Three protocol details are needed to read Table 2 honestly and are stated once here. First, the layer-wise schedule runs 200 iterations per stage for three stages, i.e. 600 Adam updates per seed, three times the per-optimiser iteration count of the others; its being the worst performer with the most updates strengthens, not weakens, the optimiser-independence conclusion. Second, SPSA was run with a gain schedule calibrated for a 2000-iteration horizon, and we read its exact loss after 200 updates from the logged `loss_exact` field (the record at iteration 201, since `loss_exact` is evaluated before each update and logged every ten iterations); the same run reaches 1.27×10^3 by iteration 2000, which we report in the text so the reader can see the 200-iteration runs are not converged. Third, iteration-matching is not compute-matching: at $L = 3$ the per-iteration loss-evaluation counts differ, giving totals of about 9,200 (finite-difference), 9,600 (layer-wise), 36,200 (parameter-shift, which evaluates a three-point time-stencil Jacobian each iteration), and ~ 425 (SPSA at iteration 200). Only finite-difference and

layer-wise are matched in evaluations (to $1.04\times$); the comparison is on a common per-iteration budget (three stages of it for layer-wise), and we make no compute-matched claim.

Optimiser (b) uses the parameter-shift Jacobian of the observables composed with the analytic Jacobian of the loss, and was verified against central finite differences. In exact simulation forward finite differencing ($\epsilon = 10^{-4}$) and exact parameter-shift feed the identical Adam loop and reproduce the same gradient, so their per-seed final losses agree to about 2×10^{-5} relative on all shared seeds; rows (a) and (b) are therefore a single gradient source measured twice, a consistency check that no finite-difference artefact is present, not two independent optimisers. The honest count is three optimiser families: gradient descent (a/b), block-coordinate layer-wise (c), and SPSA (d).

3.4. Jacobian effective dimension

For a fixed initialisation we form the exact output Jacobian $J = \partial(\langle Z_0 \rangle, \langle Z_1 \rangle, \langle Z_2 \rangle)(t)/\partial\theta$ at each of 50 collocation times by parameter-shift, stack the per-time Jacobians, and take the singular-value spectrum $\{s_i\}$ of the stacked matrix. We report two depth-resolved effective-dimension measures, each averaged over 6 random initialisations: the numerical rank at relative threshold 10^{-6} , and the participation ratio $\text{PR} = (\sum_i s_i^2)^2 / \sum_i s_i^4$, which counts the number of singular directions carrying appreciable weight. We sweep $L \in \{1, \dots, 8\}$ (that is, 15 to 120 parameters).

3.5. Fourier-expressivity ceiling

Following Schuld et al. [12], a time-encoding gate $R(\phi(t))$ with $\phi = \omega t$ and generator eigenvalues $\pm \frac{1}{2}$ contributes frequencies $\{-f, 0, +f\}$ with $f = \omega/2\pi$. The model's accessible spectrum is spanned by integer sums and differences of the active per-gate frequencies, so its maximum accessible frequency is their sum. Of the four encoding gates, the qubit-1 R_Z is inert (Section 3.1; verified numerically in Section 4.4), leaving three active frequencies: two qubits at $f = 1/t_{\max}$ (qubits 0 and 2) and one at $f = 1/2t_{\max}$ (qubit 3), with $t_{\max} = 3$ s. The accessible harmonics are therefore multiples of $1/2t_{\max}$ up to

$$f_{\max} = \frac{1 + 1 + \frac{1}{2}}{t_{\max}} = \frac{2.5}{t_{\max}} \approx 0.83 \text{ Hz}, \quad (3)$$

that is, the set $\{0, \frac{1}{2}, 1, \frac{3}{2}, 2, \frac{5}{2}\}/t_{\max}$ of 1 DC plus 5 non-zero harmonics. This is an analytic property of the fixed encoding and is independent of the trainable ansatz. To verify it numerically rather than recite an analytic constant, we sample $\langle Z_j \rangle(t)$ over a full period $[0, 2t_{\max})$ (so every accessible half-integer harmonic lands on an exact FFT bin, avoiding the spectral leakage that contaminates a half-period window) and measure the relative spectral power above $f_{\max}$ at each depth. We compare $f_{\max}$ to the 99th-percentile spectral bandwidth of the reference Lorenz trajectory, computed by FFT of each component of an RK4 solution at $\Delta t = 0.01$.

3.6. Trained depth sweep

Finally, we train the full physics loss at a matched budget (800 SPSA iterations, 10 seeds each) across $L \in \{1, 2, 3, 5, 8\}$ and report the mean and standard error of the final loss, to test whether the empirically achievable loss tracks the structural capacity ceiling identified by the Jacobian and Fourier diagnostics.

3.7. Reproducibility

All diagnostics are produced by a single checkpointed harness (`scripts/r1_experiments.py`) with recorded git revision, source-file SHA-256, and per-unit checkpoints; every result file carries its provenance block. The corrected live/dead variance split, capacity-curve dispersion, and optimiser evaluation counts are computed post hoc from those files by `scripts/analyze_r1_corrections.py`, and the corrected Fourier ceiling is verified by an independent statevector simulation in `scripts/verify_fourier_ceiling.py`; both are deterministic and carry their own SHA-256. Result files are named in the captions below.

4. Results

4.1. The gradient variance sits at the Haar floor, not on a plateau

Table 1 reports the McClean-comparable estimator of Section 3.2, split into pooled, live, and dead parameters. Three facts follow. First, the pooled mean is essentially flat as the parameter count grows fivefold (8.8×10^{-3} at $L = 1$ to 4.6×10^{-3} at $L = 5$, a factor of 1.9); taken alone this looks like the absence of any depth trend. Second, that flatness is a pooling artefact: the dead-parameter fraction falls from 86.7% ($L = 1$) to 60.0% ($L = 3$) to 37.3% ($L = 5$) as the backward light cone of Z_0 widens with depth, and the pooled mean is exactly the product of the live-only mean and the live fraction. Restricted to the live parameters, the variance decays about ninefold ($6.6 \times 10^{-2} \rightarrow 7.3 \times 10^{-3}$, a factor of 9.0). Third, and decisively, that decay is toward a floor, not toward zero: for this local cost the Haar/2-design saturation scale is $2^{-2n} = 2^{-8} = 3.9 \times 10^{-3}$ at $n = 4$ [3], and the live variance falls from 16.9 to 1.9 times this floor over $L = 1$ to $L = 5$. This is the canonical decay-then-saturate profile of a fixed- n circuit converging to its 2-design floor, and the floor itself is large enough to train. The failure is therefore not a vanished gradient. Figure 1 plots the pooled and live variances against the Haar floor and against the exponential decay a barren plateau would predict. The full per-parameter distribution of this variance across the live parameters (Figure 2) tells the same story as its mean: as depth grows the entire distribution migrates toward the Haar floor and accumulates just above it, rather than collapsing below it as a barren plateau would require.

We emphasise what a single n can and cannot show. It cannot certify or refute the n -asymptotic McClean threshold, which is inaccessible at $n = 4$. It can show

that the gradient variance sits at the 2-design floor for this ansatz and cost, and that training reduced the loss tenfold as depth grew (Section 4.6), so gradient magnitude was demonstrably not the binding constraint. The defensible statement is that gradients at $n = 4$ are at the Haar scale and are large enough to train, not that they fail to decay with depth.

Table 1. McClean-comparable gradient variance (Section 3.2): the variance across 100 random initialisations of the exact fixed-circuit, fixed- t parameter-shift gradient of the bounded local cost, on 24 collocation times. Pooling over a dead-parameter fraction that shrinks with depth makes the pooled mean look flat; the live-only mean decays about ninefold, from 16.9 to 1.9 times the local-cost Haar floor $2^{-2n} = 3.9 \times 10^{-3}$. **Source:** `results/r1/variance_b1.e1_L{1,3,5}.json` and `results/r1/r1.corrections.analysis.json` (e1.live_dead).

L	Dead frac.	Pooled mean	Live mean	Live / floor
1	86.7%	8.8×10^{-3}	6.6×10^{-2}	16.9
3	60.0%	6.4×10^{-3}	1.6×10^{-2}	4.1
5	37.3%	4.6×10^{-3}	7.3×10^{-3}	1.9

4.2. The failure is optimiser-independent

If the poor loss were an optimiser pathology, a plateau that a better gradient estimator could escape, then different optimisers should reach qualitatively different basins. They do not. Table 2 shows that at a common nominal budget of 200 optimiser iterations (600, in three stages, for layer-wise), gradient descent, layer-wise, and SPSSA optimisation all terminate within the same order of magnitude of final loss ($\sim 10^3$ – 10^4). Finite-difference and exact parameter-shift agree to about 2×10^{-5} relative per seed (their row difference is entirely the extra five finite-difference seeds), so they corroborate that no finite-difference artefact is present rather than adding a fourth independent optimiser. The layer-wise schedule, which runs three times as many Adam updates as the others (600 against 200), is the worst performer, which sharpens the conclusion. No optimiser unlocks a qualitatively better solution, consistent with a capacity ceiling and inconsistent with a gradient-estimation problem.

The one caveat is convergence: SPSSA continued to 2000 iterations in the same run reaches 1.27×10^3 , so the 200-iteration comparison establishes that the optimisers agree at a common budget, not that any of them has reached a basin floor. The qualitative conclusion (no optimiser unlocks a different order of magnitude) is unaffected, and the Jacobian and Fourier diagnostics below give the depth-resolved capacity limit directly.

4.3. The output-Jacobian effective dimension saturates with depth

Table 3 reports the effective dimension of the output Jacobian as a function of depth. The numerical rank rises through 5, 15, 25 at $L = 1, 2, 3$, reaches 32.7 at $L = 4$ (four of six

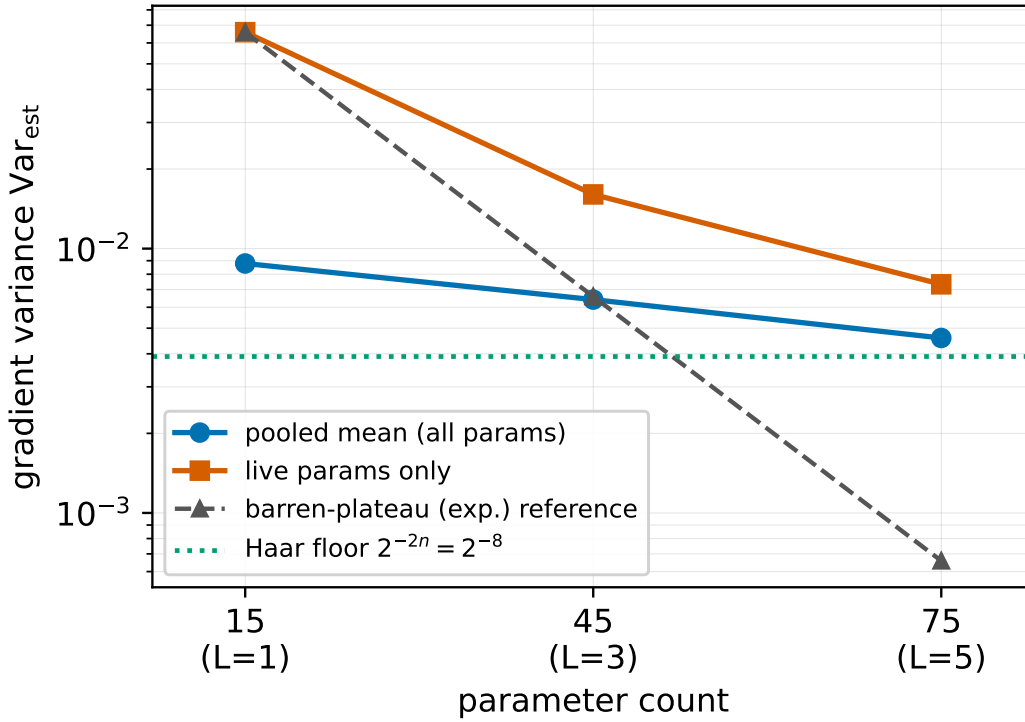


Figure 1. McClean-comparable gradient variance versus parameter count (log scale). The pooled mean (blue circles) is nearly flat, but this is a pooling artefact; the live-parameter mean (orange squares) decays about ninefold toward the local-cost Haar/2-design floor $2^{-2n} = 3.9 \times 10^{-3}$ (green dotted line) and saturates there. A barren plateau would instead fall exponentially (grey dashed reference). Source: `variance_b1_e1L{1,3,5}.json`.

Table 2. Mean final loss at a common nominal budget of 200 optimiser iterations on the $L = 3$ (45-parameter) QPINN. All land within one order of magnitude. Layer-wise runs 200 iterations per stage for three stages (600 Adam updates); finite-difference and parameter-shift agree to about 2×10^{-5} relative and are one gradient source measured twice; SPSA is read after 200 updates of a 2000-iteration-scheduled run (record iteration 201). Eval. counts are total loss evaluations at $L = 3$ (loss-evaluation equivalents for parameter-shift, per the harness cost model) and show the comparison is iteration-matched, not compute-matched. **Sources:** `train_fd_b1_e4_fd_10seed.json`, `train_ps_b1_e2b_ps.json`, `train_layerwise_b1_e2d_layerwise.json` (final_loss), `train_spsa_b1_e2a_spsa.json` (loss_exact at iteration 201).

Optimiser	Seeds	Mean final loss	Evals
Finite difference	10	1.02×10^4	9,200
Parameter-shift (exact)	5	9.51×10^3	36,200
Layer-wise (3 stages)	5	1.10×10^4	9,600
SPSA (iter. 200)	5	3.1×10^3	425

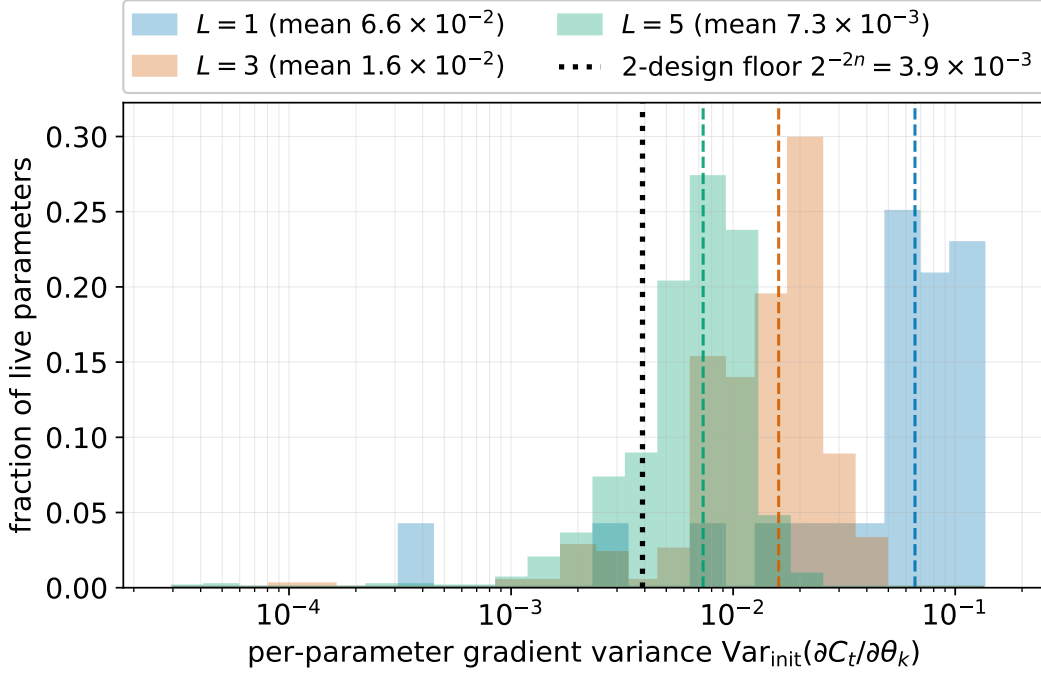


Figure 2. Distribution of the per-parameter gradient variance across the structurally live parameters at $L = 1, 3, 5$ (48, 432 and 1128 live parameter-time cells). For each live cell the plotted quantity is the variance, across the 100 random initialisations, of the exact fixed-circuit, fixed- t parameter-shift gradient of the bounded local cost; each histogram is normalised to the fraction of that depth’s live parameters. The dashed vertical lines mark the per-depth means, which coincide with the live-mean column of Table 1 (6.6×10^{-2} , 1.6×10^{-2} , 7.3×10^{-3} at $L = 1, 3, 5$). As depth grows the whole distribution shifts toward the local-cost Haar/2-design floor $2^{-2n} = 3.9 \times 10^{-3}$ (black dotted line) and accumulates just above it, the decay-then-saturate signature rather than an exponential collapse. Source: `variance_b1_e1_L{1,3,5}.json`.

initialisations give 33, two give 32), and is then flat at exactly 33 across all initialisations from $L = 5$ through $L = 8$: adding the 45 parameters from $L = 5$ to $L = 8$ produces zero new independent output directions. The participation ratio continues to rise ($6.8 \rightarrow 9.2$ over $L = 4$ to $L = 8$), so the added parameters redistribute weight among directions that already exist without creating new ones. This is a directly measured capacity ceiling in the space of functions the circuit’s output can represent. Section 4.5 shows that the ceiling value 33 is exactly the dimension of the corrected band-limited function space.

4.4. A depth-independent Fourier-expressivity ceiling

Why does the Jacobian rank saturate? Because the circuit is structurally band-limited by its encoding. The accessible maximum frequency of Equation (3) is $f_{\max} = 2.5/t_{\max} \approx 0.83$ Hz, set by the three active encoding gates and independent of the trainable ansatz. We verify this numerically rather than reciting an analytic constant. First, deleting the inert qubit-1 R_Z encoding changes $\langle Z_0 \rangle, \langle Z_1 \rangle, \langle Z_2 \rangle$ by at most 1.6×10^{-15} at every

Table 3. Effective dimension of the output Jacobian $\partial(\langle Z_0 \rangle, \langle Z_1 \rangle, \langle Z_2 \rangle)/\partial\theta$ versus depth, averaged over 6 initialisations. The numerical rank saturates at 33 from $L = 5$ onward (mean 32.7 at $L = 4$) while the parameter count keeps growing. **Source:** `results/r1/jacobian.b1.e3.jac.L{1..8}.json`, per-entry `rank_1e-6` and `participation_ratio`.

L	Params	Rank (10^{-6})	Participation ratio
1	15	5	2.0
2	30	15	3.5
3	45	25	6.2
4	60	32.7	6.8
5	75	33	7.7
6	90	33	8.2
7	105	33	8.8
8	120	33	9.2

depth $L \in \{1, 3, 5, 8\}$, confirming it contributes no frequency. Second, sampling the observables over the full period $[0, 2t_{\max})$ and taking the FFT on the commensurate grid, the relative spectral power above $f_{\max}$ is at machine precision ($\sim 10^{-30}$) for every observable and every depth; nothing lies above the ceiling. (An earlier random-draw spectrum sampled on a half-period window showed apparent power above the ceiling; that was spectral leakage from the half-integer harmonics falling between FFT bins, and it vanishes on the full-period grid.) The band edge $f_{\max}$ itself becomes populated once depth is sufficient to route the qubit-3 half-frequency through the entangler onto the Z_0 cost observable, which is exactly the depth dependence the rank column records.

Table 4 compares the ceiling to the Lorenz target bandwidths. Even the narrowest component (the x -component, 99th-percentile bandwidth 3.65 Hz) exceeds the accessible ceiling by a factor of 4.4; the y and z components (4.32 and 15.61 Hz) exceed it by factors of 5.2 and 18.7. Expressed as target spectral energy, 36% of the x -component’s energy, 54% of the y -component’s, and 85% of the z -component’s lies above the ceiling and is therefore unreachable by the circuit at any depth. Adding variational layers reparametrises energy within the accessible band without widening it, which is the mechanistic cause of the Jacobian-rank saturation in Section 4.3: more depth cannot create output directions at frequencies the encoding cannot reach.

4.5. The band dimension equals the measured rank ceiling

The Fourier ceiling and the Jacobian rank are the same result seen two ways. The accessible band is $\{0, \frac{1}{2}, 1, \frac{3}{2}, 2, \frac{5}{2}\}/t_{\max}$: one DC component plus five non-zero harmonics, each contributing a cosine and a sine, so $1+2 \times 5 = 11$ real basis functions per observable. With three observables $\langle Z_0 \rangle, \langle Z_1 \rangle, \langle Z_2 \rangle$ the reachable output functions span at most

$$3 \times (1 + 2 \times 5) = 3 \times 11 = 33 \quad (4)$$

Table 4. Maximum accessible output frequency and the Lorenz target bandwidths. The accessible frequency is set by the fixed encoding, is independent of depth, and lies a factor of 4.4–18.7 below the per-component target bandwidths; the final column is the fraction of that component’s spectral energy above the ceiling. **Source:** `results/r1/fourier_ceiling_verification.json` (ceiling, depth independence) and `results/r1/fourier_b1.e3_fourier_L1.json` (`target_bandwidth_99pct_hz`, `target_spectrum`).

Component	Target bw. (Hz)	Gap to $f_{\max}$	Energy above
x	3.65	$4.4\times$	36%
y	4.32	$5.2\times$	54%
z	15.61	$18.7\times$	85%
$f_{\max} = 2.5/t_{\max} \approx 0.83$ Hz, identical at $L \in \{1, 3, 5, 8\}$.			

dimensions, which is exactly the measured rank ceiling of Table 3. Two independent diagnostics, an analytic encoding-frequency count and a numerical singular-value rank, therefore agree on the single number 33. This closure is the strongest single result of the paper: it identifies the capacity ceiling with a countable set of accessible Fourier modes and confirms it against a directly measured rank. We note that the earlier, incorrect ceiling $3.5/t_{\max}$ would have implied $3 \times (1 + 2 \times 7) = 45$ basis functions, which cannot explain a rank ceiling of 33; the corrected ceiling closes the chain and the incorrect one does not.

The closure is exact only at saturation. The intermediate ranks 5, 15, 25 at $L = 1, 2, 3$ are not a single $3 \times (1 + 2m)$ formula, because the reachable harmonic set grows with depth and differs across the three observables at low depth. At $L = 1$ only the integer harmonics $\{0, 1, 2\}/t_{\max}$ are reachable and they are split across observables ($\langle Z_0 \rangle, \langle Z_1 \rangle$ each reach $\{1\}$, $\langle Z_2 \rangle$ reaches $\{0, 2\}$), so the per-observable block ranks are 2, 2, 3 and support counting bounds the stacked rank by $2 + 2 + 3 = 7$; two cross-observable linear dependencies among the shared low-depth parameter directions reduce this to the measured stacked rank of 5. The qubit-3 half-integer harmonics become reachable through the entangler only at $L \geq 2$, the full band is reached on all three observables at $L = 4$ (with the cost observable $\langle Z_0 \rangle$ lagging the others at $L = 3$), and the every-initialisation rank saturation at 33 completes one layer later at $L = 5$ (32.7 on average at $L = 4$). The corrected ceiling $f_{\max}$ is exact at every depth (there is never power above it); it is the filling of the band, not the ceiling, that grows with depth.

4.6. The trained loss is flat once the rank has saturated

The structural diagnostics predict an empirical signature: trained loss should improve with depth up to the point where the Jacobian rank saturates ($L \approx 4$ –5), then stop improving. Table 5 is consistent with this. At a matched 800-iteration SPSA budget the mean final loss falls from $L = 1$ to $L = 5$ and is flat from $L = 5$ to $L = 8$ (1.25×10^3

against 1.19×10^3). We report standard errors over the ten seeds because they qualify the strength of the claim. The dispersion is large at shallow depth (standard errors of 1294, 772, 355 at $L = 1, 2, 3$) and small once the loss has plateaued (55, 104 at $L = 5, 8$). The improvement from $L = 3$ (2246 ± 355) to $L = 5$ (1250 ± 55) is about 2.8 combined standard errors, so the fall onto the plateau is resolved; $L = 4$, where the rank actually saturates, was not in the trained grid, so the precise onset location is bracketed rather than pinpointed. The data are thus consistent with the loss plateauing where the rank saturates, and we state the coincidence as consistency at the resolution of the grid, not as an exactly located transition. The robust part of the signal, monotone improvement over $L = 1-3$ and a flat plateau over $L = 5-8$, matches the capacity picture.

Table 5. Mean final loss (with standard error over 10 seeds) versus depth at a matched 800-iteration SPSA budget. Loss improves with depth up to $L = 5$ and is then flat, consistent with the Jacobian-rank saturation of Table 3; the shallow-depth spread is large, so the onset is reported as consistency rather than an exactly located transition. **Source:** `results/r1/train_spsa_b1.e3_cap_L{1,2,3,5,8}.json` and `results/r1/r1_corrections_analysis.json` (capacity_dispersion).

L	Params	Mean final loss (SEM)
1	15	9.82×10^3 (1.3×10^3)
2	30	4.39×10^3 (7.7×10^2)
3	45	2.25×10^3 (3.6×10^2)
5	75	1.25×10^3 (5.5×10^1)
8	120	1.19×10^3 (1.0×10^2)

The training dynamics themselves (Figure 3) show the oscillatory, high-across-seed-variance behaviour of an under-parameterised fit with a live gradient, not the flat, stalled signature of a vanished gradient.

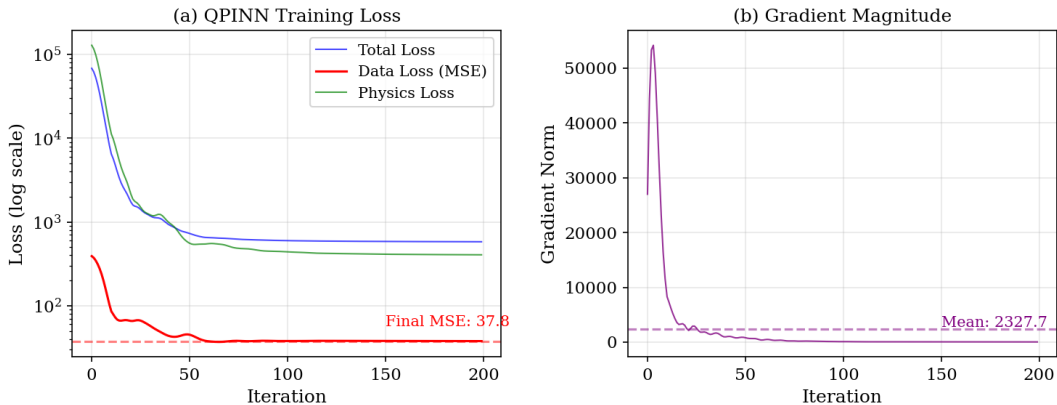


Figure 3. Representative QPINN training dynamics: the loss oscillates and plateaus while the gradient remains large throughout, the signature of an under-parameterised fit rather than of a barren plateau. (Reproduced from the setup of [7].)

5. Discussion

5.1. Correcting our earlier barren-plateau diagnosis

An earlier, unpublished version of this analysis reported that the QPINN was not on a barren plateau on the basis that its measured gradient norms remained in the range 10^3 – 10^4 throughout training, four to five orders of magnitude above the McClean threshold. That argument reached a defensible conclusion for the wrong reason, and we correct the measurement here. The gradient norm of the time-averaged physics loss is not a McClean-comparable quantity: it carries the arbitrary scale of the unbounded, unnormalised loss, and averaging over the encoding period before differentiating annihilates the very Fourier modes the McClean bound concerns. Comparing an unnormalised $\|\nabla\mathcal{L}\|$ of order 10^3 to a dimensionless 2^{-n} threshold is a category error.

The corrected measurement (Section 4.1) fixes this. It uses (a) a bounded local cost $C_t \in [0, 1]$, removing the arbitrary scale; (b) a fixed circuit at a fixed time, so no time-averaging destroys the relevant modes; and (c) exact parameter-shift gradients, removing any finite-difference artefact. On this quantity the variance sits at the local-cost Haar/2-design floor $2^{-2n} = 3.9 \times 10^{-3}$, and on its live parameters it decays toward that floor with depth rather than falling to zero. The earlier four-to-five-orders cushion was an artefact of the unnormalised loss scale and does not survive normalisation: after normalisation the variance is at the 2-design floor, not far above it. The conclusion (no vanished gradient) is unchanged; the magnitude of the claim is much more modest, and the correct reading is that gradient magnitude is not the binding constraint, capacity is.

We also do not repeat the earlier framing that presented this disambiguation as a novel general-purpose diagnostic. The contribution here is narrower and more defensible: for this specific circuit and task, four measurements agree that the ceiling is capacity/expressivity, not a plateau.

5.2. Why depth cannot help: encoding-limited expressivity

The four diagnostics form a single mechanistic chain. The encoding fixes the accessible frequency band at $2.5/t_{\max}$ (Section 4.4); the band is narrower than the target, so the reachable output functions span a fixed space of dimension 33; the Jacobian rank therefore saturates at 33 once depth is sufficient to populate that space (Sections 4.3 and 4.5); the trained loss is then flat at the same depth (Section 4.6); and throughout, the gradient variance sits at the Haar floor and is large enough to train (Section 4.1), with no optimiser doing better (Section 4.2). The bottleneck is the fixed encoding, not the number of trainable parameters and not the optimiser. This is a concrete instance of the Schuld et al. [12] result that the encoding sets the model’s Fourier support: to fit a higher-bandwidth target one must change the encoding (for example richer data re-uploading), not merely add layers.

5.3. Architectural, not quantum: the reservoir and ESN comparison

On the same Lorenz task, a fixed quantum reservoir with a closed-form ridge readout and a purely classical echo-state network (ESN) both outperform the variational QPINN by a wide margin, and in a fair comparison the tuned classical ESN is the strongest of the three [7, 20]. It would be tempting to read the reservoir’s advantage as a quantum effect; the diagnosis here shows it is not. The relevant contrast is architectural: a fixed, high-dimensional random feature map with a closed-form linear readout against a small, band-limited, end-to-end-trained circuit. The former avoids the capacity ceiling by supplying a rich fixed feature basis and solving the readout in closed form (Figure 4); the latter is limited by the fixed encoding’s narrow Fourier support and by its handful of trainable parameters. Because both the classical ESN and the quantum reservoir enjoy the architectural advantage, the advantage is not quantum mechanical. Our companion conditioning analysis [19] makes the same point from the numerical-stability side: the fixed random feature map has a bounded condition number where the trained alternative does not, so the architectural gap is visible in the conditioning of the readout problem independently of any quantum consideration.

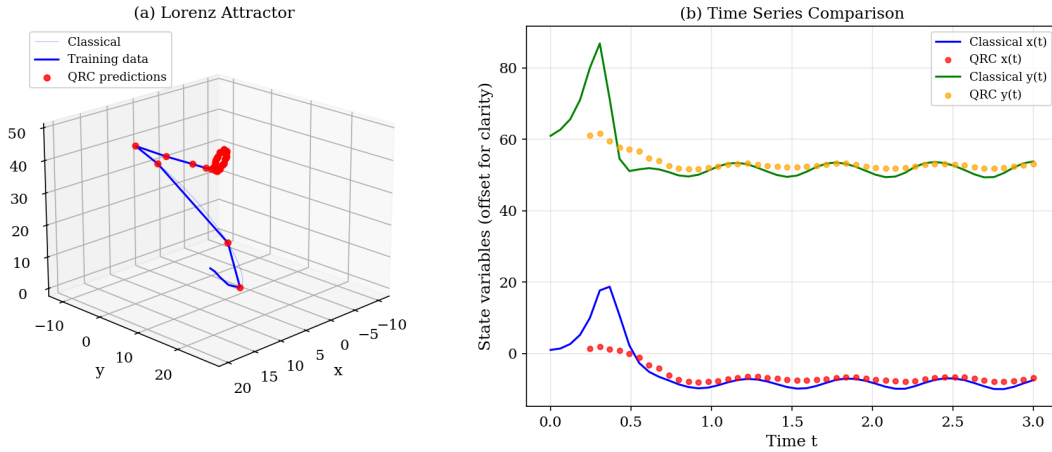


Figure 4. The fixed high-dimensional reservoir tracks the chaotic trajectory that the band-limited variational circuit cannot. Quantum-reservoir (QRC) predictions with a closed-form ridge readout, overlaid on the reference Lorenz-63 solution. (a) 3D attractor view: the QRC predictions (red) follow the reference orbit. (b) Component-wise time series (offset for clarity): QRC $x(t)$ and $y(t)$ track the reference to within marker width. The fixed random feature basis supplies enough capacity to reconstruct the orbit, whereas the fixed-encoding variational circuit is capped by its narrow Fourier support (Section 4.4); the contrast is architectural rather than quantum.

The summary is therefore modest and specific. We do not claim a quantum advantage; we give a mechanistic explanation for a widely observed variational-circuit failure, and we locate the observed performance differences in architecture rather than in quantum physics.

5.4. Scope and limitations

We deliberately do not make near-term-hardware claims. The study is a four-qubit, exactly simulated diagnosis; the value of the small scale is that every diagnostic is exact, so the barren-plateau question can be settled without sampling artefacts, not that the result is a device demonstration. Three limitations follow. First, the barren-plateau statement is a scaling claim in the qubit number, which a single $n = 4$ experiment cannot settle in the asymptotic sense; what we establish is that at this width the gradient is at the Haar floor and is large enough to train, so the observed failure is not gradient-magnitude-limited. Second, the encoding-set frequency ceiling is a property of this encoding; a different encoding (denser data re-uploading, learnable encoding frequencies) would move the ceiling, and testing that is the natural next experiment. Third, we have shown that depth does not help this architecture on this target; we have not shown that no small variational circuit can fit chaotic dynamics, only that fitting requires widening the accessible band, which is an encoding choice rather than a depth choice.

6. Conclusion

The failure of a four-qubit variational quantum circuit to fit the Lorenz-63 attractor is not a barren plateau. A McClean-comparable, fixed-circuit, exact-parameter-shift gradient variance sits at the local-cost Haar floor $2^{-2n} = 3.9 \times 10^{-3}$ and, on its live parameters, decays toward that floor with depth rather than collapsing to zero; three optimiser families reach the same order of magnitude of loss at matched budget; the output-Jacobian rank saturates at 33 from five layers on; and the accessible output frequency band is fixed by the encoding at $2.5/t_{\max} \approx 0.83$ Hz, a factor of about 4.4 below even the narrowest target component, at every depth. The corrected band has dimension $3 \times (1 + 2 \times 5) = 33$, exactly the measured rank ceiling, tying the Fourier and Jacobian diagnostics into one number. These measurements agree on a single mechanism: an architectural capacity ceiling set by the fixed encoding, not a trainability plateau. We correct our earlier, unnormalised gradient-norm diagnosis, and we place the observed advantage of fixed reservoirs and classical echo-state networks on this task in architecture rather than in quantum mechanics.

Acknowledgments

The author used Claude (Anthropic) to assist with running and optimizing the simulation code. All technical content, results, and conclusions are the author's own. This work received no external funding. The author declares no conflicts of interest.

Data and code availability

All diagnostic code and result files (with recorded git revision and per-file SHA-256 provenance) are available at https://github.com/pandey-tushar/Quantum_Chaos_solver. The diagnostics of Section 4 are produced by `scripts/r1_experiments.py`; the corrected live/dead variance split and capacity dispersion are computed by `scripts/analyze_r1_corrections.py`, and the corrected Fourier ceiling by `scripts/verify_fourier_ceiling.py`. The result files are named in each table caption.

- [1] J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush, and H. Neven, Barren plateaus in quantum neural network training landscapes, *Nat. Commun.* **9**, 4812 (2018).
- [2] M. Cerezo *et al.*, Variational quantum algorithms, *Nat. Rev. Phys.* **3**, 625–644 (2021).
- [3] M. Cerezo, A. Sone, T. Volkoff, L. Cincio, and P. J. Coles, Cost function dependent barren plateaus in shallow parametrized quantum circuits, *Nat. Commun.* **12**, 1791 (2021).
- [4] A. V. Uvarov and J. D. Biamonte, On barren plateaus and cost function locality in variational quantum algorithms, *J. Phys. A: Math. Theor.* **54**, 245301 (2021).
- [5] C. Ortiz Marrero, M. Kieferová, and N. Wiebe, Entanglement-induced barren plateaus, *PRX Quantum* **2**, 040316 (2021).
- [6] J. Qi, C.-H. H. Yang, P.-Y. Chen, and M.-H. Hsieh, TensorHyper-VQC: a tensor-train-guided hypernetwork for robust and scalable variational quantum computing, *npj Quantum Inf.* **12**, 70 (2026).
- [7] T. Pandey, Quantum reservoir computing versus variational quantum physics-informed networks for chaotic forecasting, arXiv:2604.23743.
- [8] E. N. Lorenz, Deterministic nonperiodic flow, *J. Atmos. Sci.* **20**, 130–141 (1963).
- [9] M. Raissi, P. Perdikaris, and G. E. Karniadakis, Physics-informed neural networks, *J. Comput. Phys.* **378**, 686–707 (2019).
- [10] A. Pattnaik *et al.*, Quantum-classical physics-informed neural networks for solving nonlinear differential equations, arXiv:2503.16678 (2025).
- [11] Y.-C. Leung *et al.*, Quantum physics-informed neural networks for solving differential equations, arXiv:2212.00360 (2022).
- [12] M. Schuld, R. Sweke, and J. J. Meyer, Effect of data encoding on the expressive power of variational quantum-machine-learning models, *Phys. Rev. A* **103**, 032430 (2021).
- [13] H. Jaeger and H. Haas, Harnessing nonlinearity: predicting chaotic systems and saving energy in wireless communication, *Science* **304**, 78–80 (2004).
- [14] J. Pathak, B. Hunt, M. Girvan, Z. Lu, and E. Ott, Model-free prediction of large spatiotemporally chaotic systems from data, *Phys. Rev. Lett.* **120**, 024102 (2018).
- [15] K. Fujii and K. Nakajima, Harnessing disordered-ensemble quantum dynamics for machine learning, *Phys. Rev. Applied* **8**, 024030 (2017).
- [16] P. Mújal *et al.*, Opportunities in quantum reservoir computing and extreme learning machines, *Adv. Quantum Technol.* **4**, 2100027 (2021).
- [17] K. Nakajima, K. Fujii, M. Negoro, K. Mitarai, and M. Kitagawa, Boosting computational power through spatial multiplexing in quantum reservoir computing, *Phys. Rev. Applied* **11**, 034021 (2019).
- [18] S. Ahmed, L. Tennie, and L. Magri, Robust quantum reservoir computing for chaotic dynamics, *Proc. R. Soc. A* **481**, 20250550 (2025).
- [19] T. Pandey, A quantum reservoir architecture for chaotic forecasting and a test of whether its high dimension helps, arXiv:2607.07978 (2026).
- [20] T. Pandey, When classical baselines are tuned as carefully as the quantum model, does quantum reservoir computing still win?, arXiv:2607.09905 (2026).